Fundamental Theorem of Calculus

Exam code: 9MA0

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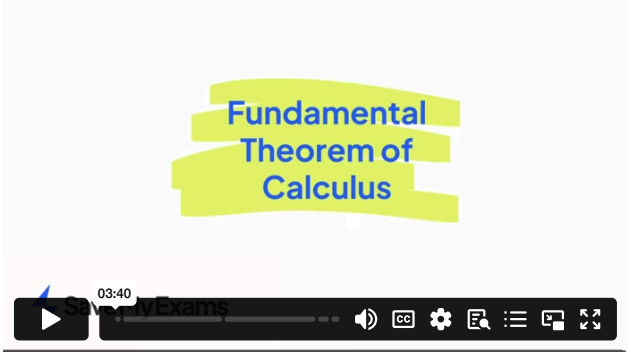

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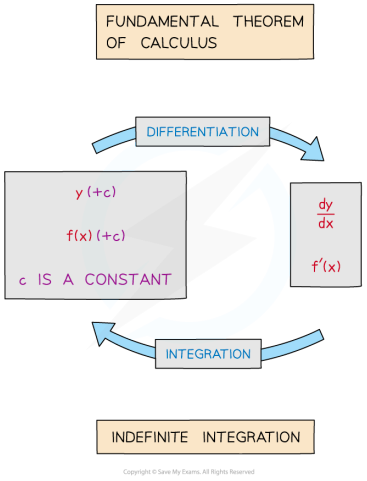
Help me understand

Fundamental theorem of calculus

What is the fundamental theorem of calculus?



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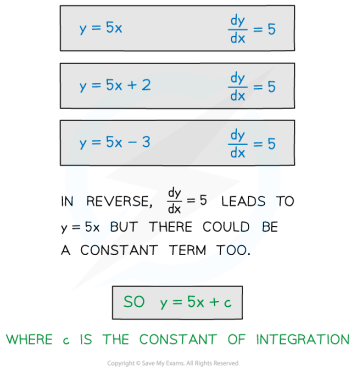
Ask questions and get help with exactly what you're revising.
Got it

Integration

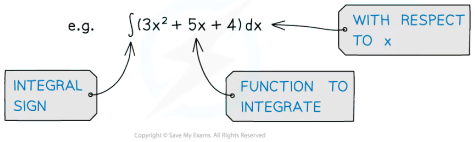
- An alternative version of the Fundamental Theorem of Calculus involves **Definite Integration**

What is a constant of integration?

- When differentiating **y**, constant terms 'disappear'
  - for constants **y = c**, $\frac{dy}{dx} = 0$
  - graphs of constants are horizontal lines and so have gradient $\left(\frac{dy}{dx}\right)$ of 0
- Integrating $\frac{dy}{dx}$, to get **y**, cannot determine the constant
  - To acknowledge this constant, "**+ c**" is used
  - **c** is called the **constant of integration**



What notation is used in integration?



- $\int$ is the sign for integration
- If it has more than one term the function to be integrated (called the **integrand**) should be in brackets
  - "Integrate" -- "**all** of (...)" -- "with respect to x"
- **dx** means integrate with respect to **x**, any other letter is treated like a number (ie like a constant)

Worked Example

(a) Find $\frac{dy}{dx}$ for $y = 6x + 3$

(b) Find $\int 6 \, dx$

a) $\frac{dy}{dx} = 6$

b) $y = 6x + c$

CONSTANT OF INTEGRATION

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